



Pacific Atlantic Water Flow / Two Stations Matrix Problem

PART 4 — Permanent DSA Pattern Sheet (Interview Revision)



Pattern Name

Reverse Flow + Multi-Source DFS/BFS

Also Known As

Reverse Graph Traversal

Multi-Source DFS

Reverse Reachability

Ocean Flow Pattern

The Core Idea

Instead of asking

Can every node
reach Destination?

Ask

Can Destination
discover every node
that can reach it?

This single question converts

$O((n*m)^2)$

↓

 $O(n*m)$

The Pattern

Suppose interview asks

Find all cells
that can reach X.

Most beginners think

Every Cell

↓

DFS

↓

Reach X?

Wrong direction.

Instead think

X

↓

Reverse DFS

↓

Find every reachable cell.

Recognition Pattern

Whenever you read

Can reach...

Immediately ask yourself

Can I reverse the search?

Recognition Keywords

These words should trigger this pattern.

Reach

Flow

Spread

Propagate

Transmit

Travel

Escape

Drain

Water

Signal

Ocean

Boundary

Exit

Border

Destination

Can reach

Eventually reach

Connected to

Recognition Checklist

While reading the problem ask

1.

Am I checking
whether every node
reaches one destination?

YES

↓

Go Next.

2.

Will DFS from every node
repeat work?

YES

↓

Go Next.

3.

Can I start
from the destination?

YES

↓

Go Next.

4.

Can I reverse
the movement rule?

YES

↓

Reverse DFS/BFS.

Biggest Observation

Brute Force

Cell

↓

Destination?

Optimization

Destination

↓

Cell

Nothing else changes.

Reverse Flow Rule

Original

Neighbour $\leq$ Current

Reverse

```
Neighbour >= Current
```

General Formula

```
Reverse
```

```
=
```

```
Reverse every edge.
```

Important Interview Question

Interviewer

```
Why >= ?
```

Answer

```
Because during reverse DFS  
I'm asking
```

```
"Could this neighbour  
have reached me  
in the original graph?"
```

```
If yes,
```

```
I move there.
```

Never say

```
Because reverse flow.
```

Explain WHY.

Interview Thinking Framework

Always think in this order.

Step 1

Understand movement.

```
How can I move?
```

Step 2

Find destination.

```
Where am I trying to reach?
```

Step 3

Brute Force

```
Run DFS  
  
from every node.
```

Step 4

Observe repetition.

```
Same cells  
  
visited  
  
again and again.
```

Step 5

Reverse the question.

Instead of

```
Can node reach destination?
```

Think

Can destination
find all nodes?

Step 6

Reverse movement.

<=
↓
>=

Step 7

Run Multi-Source DFS.

Step 8

Collect answer.

Generic Algorithm Template

Destination Cells

↓

Push into DFS/BFS

↓

Reverse Movement

↓

Mark Reachable

↓

Repeat for every destination

↓

Intersection

↓

Answer

Universal Code Template

```
visited=[[False]*m for _ in range(n)]

def dfs(r,c):

    visited[r][c]=True

    for neighbour:

        if valid:

            if visited:

                continue

            if reverse_condition:

                dfs(neighbour)
```

Then

```
for every source:

    dfs(source)
```

Done.

How to Identify Reverse Graph Problems

If interviewer asks

Can every node
reach one node?

Think

Reverse Graph.

If interviewer asks

Can every city
reach capital?

Think

Start from capital.

If interviewer asks

Can every room
reach exit?

Think

Start from exit.

If interviewer asks

Can every student
reach principal office?

Think

Start from office.

Real World Analogies

Example 1

Bus Stop

Question

Which houses
can reach Bus Stop?

Wrong

House1

↓

Bus Stop

House2

↓

Bus Stop

Correct

Bus Stop

↓

Walk backwards

↓

Find every house.

Example 2

River

Question

Which villages
can send water
to river?

Wrong

Village

↓

River

Correct

River

↓

Move uphill

↓

Find villages.

Example 3

Internet

Question

Which computers
can send packets
to server?

Wrong

Computer

↓

Server

Correct

Server

↓

Reverse connections

↓

Find computers.

Why Multi-Source DFS?

Pacific is NOT

One Cell.

Pacific is

Entire Boundary.

Therefore

Sources

Top Row

+

Left Column

Similarly

Atlantic

Bottom Row

+

Right Column

Whenever destination has

Many starting points

↓

Multi-Source DFS/BFS

should click.

Common Mistakes

Mistake 1

Memorizing

$\geq$

Instead derive it.

Mistake 2

Thinking

Water reversed.

Wrong.

Search reversed.

Mistake 3

Starting DFS

from every cell.

Mistake 4

Forgetting

Visited Matrix.

Mistake 5

Running DFS

without reverse condition.

Similar Interview Problems

1. Pacific Atlantic Water Flow (LeetCode 417)

Exactly this pattern.

2. Surrounded Regions (LeetCode 130)

Border

↓

Reverse DFS

↓

Mark Safe Cells

Pattern

Reverse Boundary DFS

3. Number of Enclaves (LeetCode 1020)

Boundary

↓

Remove reachable land

↓

Remaining answer.

Same idea.

4. Walls and Gates (LeetCode 286)

All Gates

↓

Multi-Source BFS

↓

Fill Distances.

5. Rotting Oranges (LeetCode 994)

All Rotten

↓

Multi-Source BFS.

6. 01 Matrix (LeetCode 542)

All Zeroes

↓

Multi-Source BFS.

7. Escape Large Maze

Reverse thinking appears.

8. Reverse Graph Reachability

Any graph problem asking

Who can reach destination?

Pattern Family

Grid DFS

↓

Boundary DFS

↓

Reverse DFS

↓

Multi Source DFS

↓

Reverse Graph

These are all closely related.

Complexity Pattern

Brute Force

Nodes

×

DFS

=

$O(V^2)$

Reverse DFS

One Traversal

=

$O(V)$

Grid

$O(n*m)$

Questions to Ask Yourself During Interview

1.

Am I starting DFS
from every node?

2.

Am I repeating work?

3.

Can I reverse
the search?

4.

Can destination
be my starting point?

5.

Do I have
multiple destinations?

↓

Multi-Source DFS?

6.

What happens
to movement rule
after reversing?

One-Line Recognition Rule

Node → Destination ?

↓

Think

Destination → Node

30-Second Revision Sheet

Problem

↓

Can node reach destination?

↓

Brute Force

DFS from every node

↓

Repeated work

↓

Reverse Search

↓

Start from destination

↓

Reverse movement

↓

Multi-Source DFS

↓

Visited Matrix

↓

Intersection

↓

Answer

10-Second Interview Cheat Sheet

Destination?

↓

Reverse DFS

↓

>=

↓

Multi Source

↓

Visited

↓

Intersection

Master Formula

Whenever you see

Can every node
reach X?

Immediately think

Start from X
↓
Reverse DFS/BFS
↓
Mark all reachable nodes
↓
Done.

Final Takeaway

This problem is **not about matrices**.

It is **not about water**.

It is **not about DFS**.

The real interview pattern is:

Whenever many nodes independently ask "Can I reach the same destination?", consider reversing the search. Start from the destination (or all destination boundary cells), traverse the graph in reverse, and mark every node that can reach it.

Once this pattern clicks, you'll recognize it in many graph and grid problems—not just Pacific Atlantic Water Flow.